

projection geometry: single-guide instance (room wall + speed ball)

region: feasible set (half-space intersect speed ball, room-wall constraint

blue, $\|u\| \leq v_{\max}$ dashed). Arrow: projection of u_{nom} onto the set; purple arrows are

active-constraint normals. Dykstra output coincides with the 50-digit reference within $1e-6$ m/s.

ication: Numerically cross-checked at 50-digit precision (tested instance)

745582d849 | Wolfram 15.0.0 | WorkingPrecision 50 | cross-language tol $1e-9$ | projection distance $|u_{\text{dykstra}} - u_{\text{ref}}| \leq 1e-6$ m/s

